Project 3

Lecturer: Shuai Wang

TA: Hao Gong

Requirements for submitting the assignment

- **Email address：seudatabaseqa@seu.edu.cn**
- **Format of Email subject and File name:**

  StudentNumber-Name-Project-X

  example : *71121000-张三-Project-1.zip*

- **Including：**

  Submit only necessary code files such as **java/xml** etc. Other files that you think are important can also be included (such as README). The remaining project files do not need to be submitted.
  Please limit the size of your code package.

  ① *report_A3.doc / report_A3.docx / report_A3.pdf*

  ② *code.zip*

Deadline: Beijing time, 11.15, 23:59:59

Project 3

I. Tutorial with Project Examples

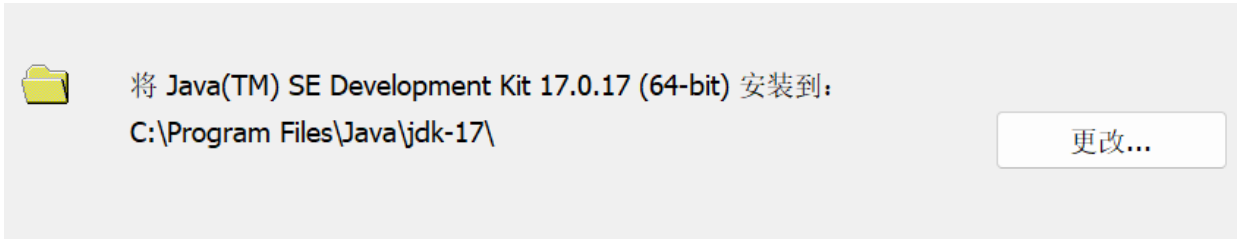
II. Assignment

Tutorial with Project Examples

1. Preparation Work

Download JDK17 for Microsoft Windows

- **Install JDK17** jdk-17.0.17_windows-x64_bin.exe
- **Change path (default path contains spaces, not recommended)**



- **Installation completed**

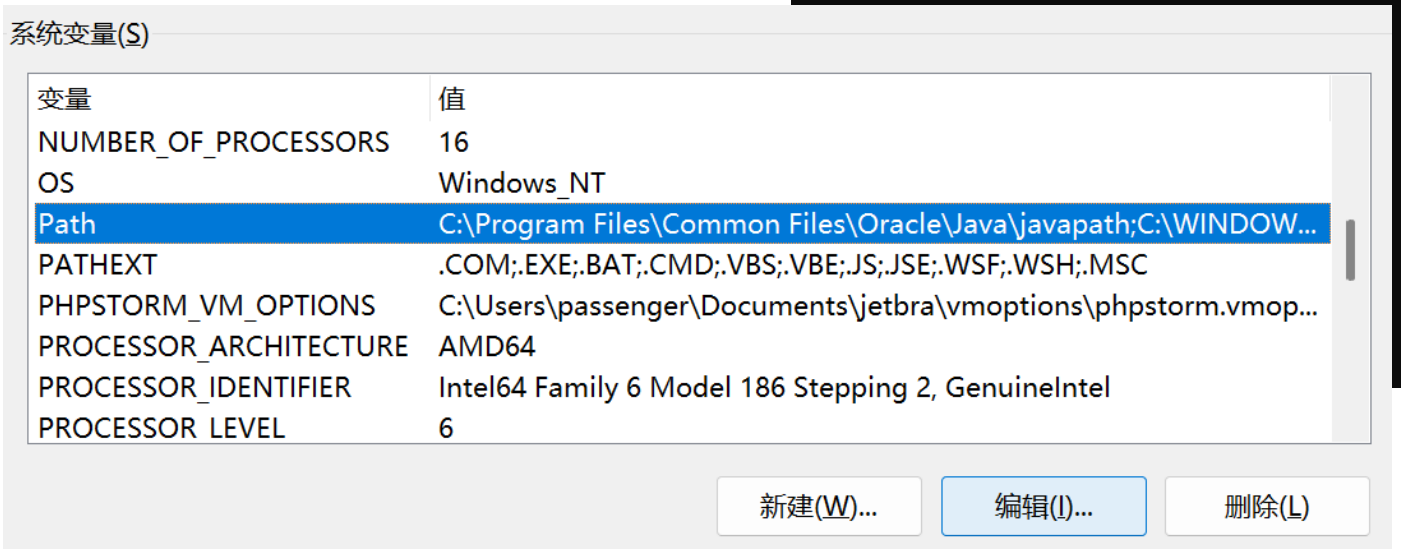


Tutorial with Project Examples

- **Press Win+R, enter sysdm.cpl → Advanced → Environment Variables**

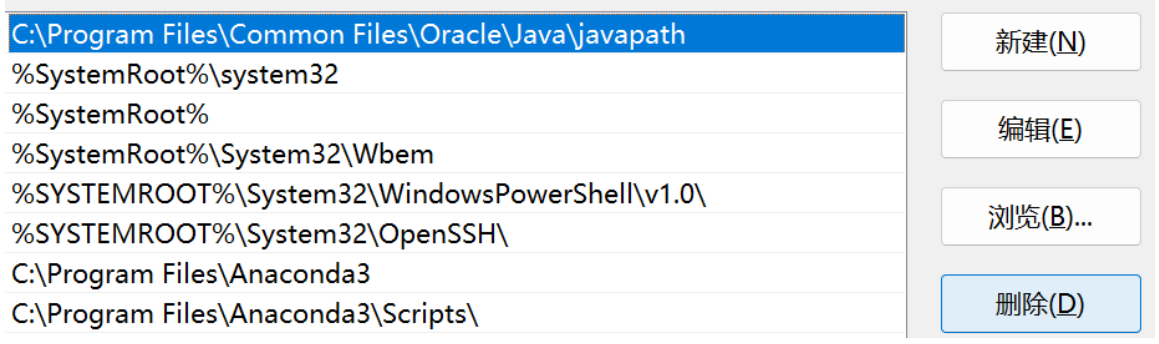


- **Edit system variables**

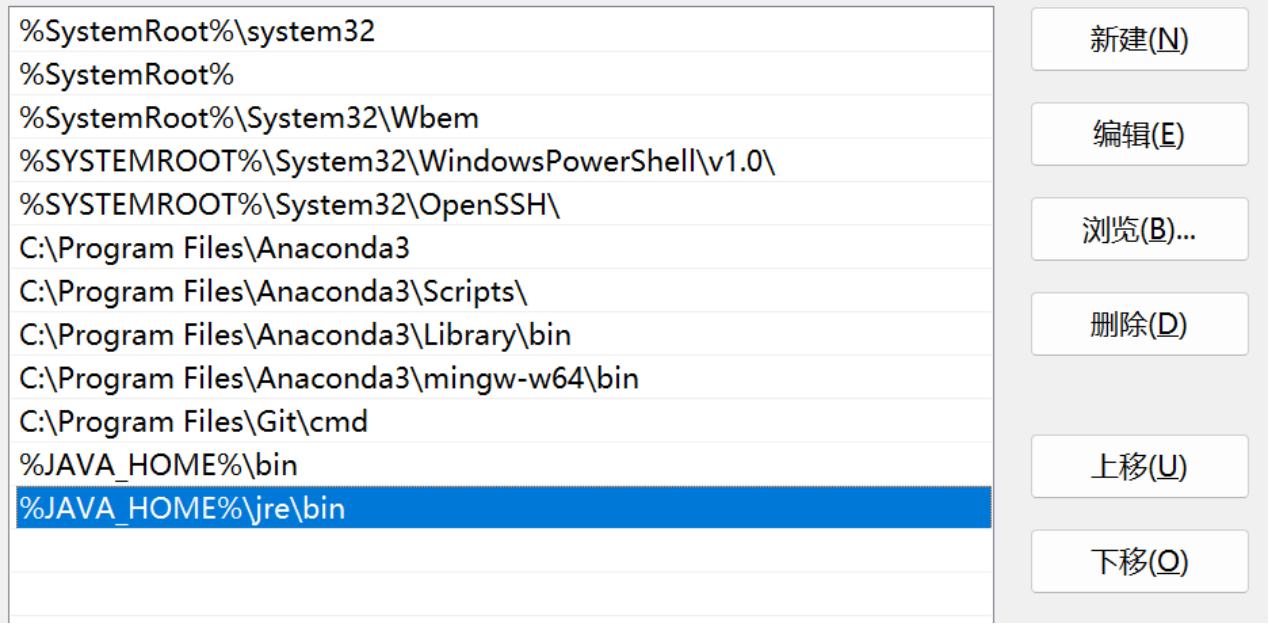


Tutorial with Project Examples

- **Delete**

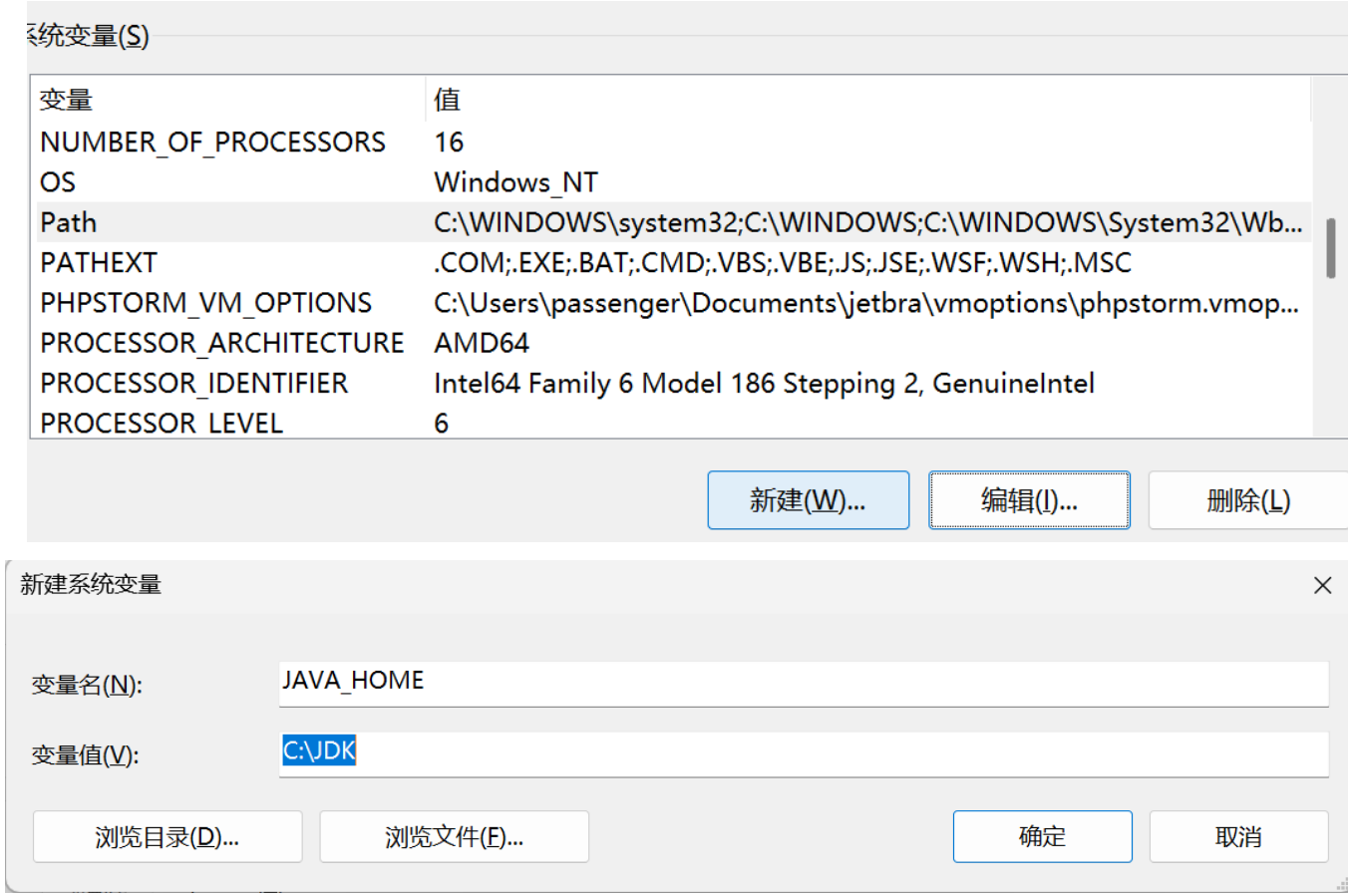


- **Add two variables in Path %JAVA_HOME%\bin and %JAVA_HOME%\jre\bin**



Tutorial with Project Examples

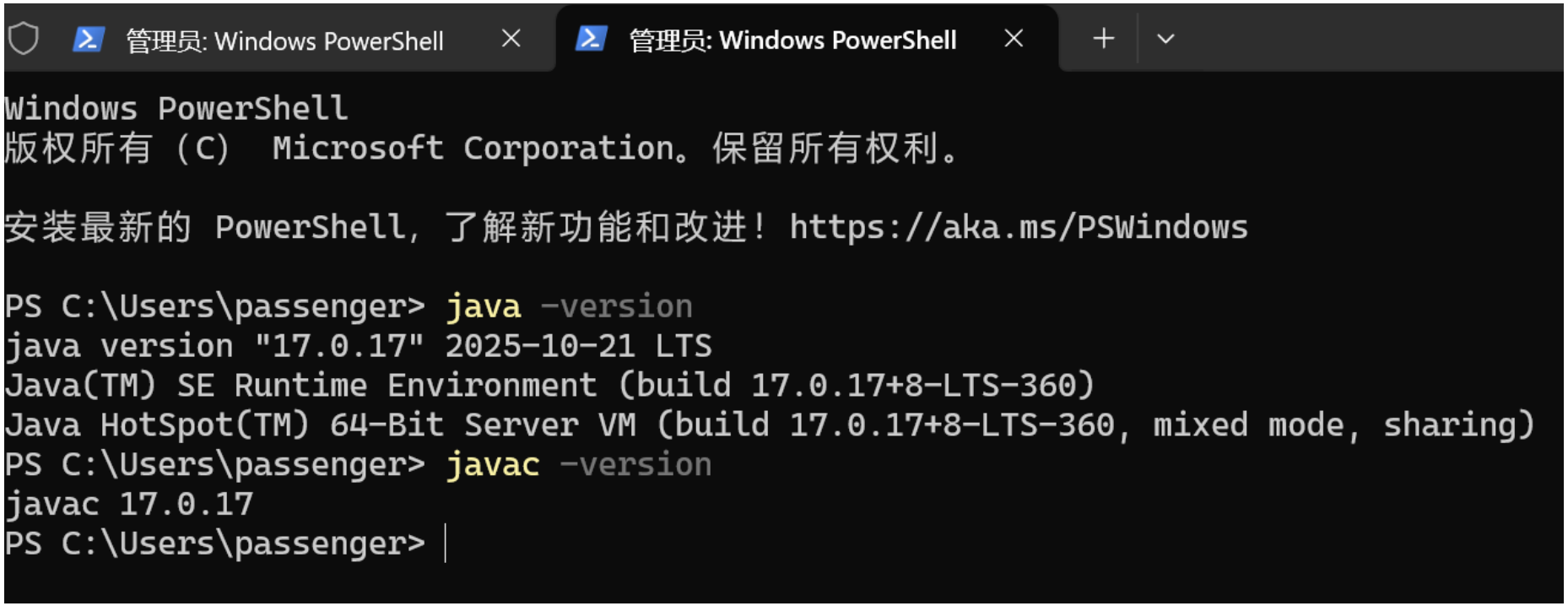
- **Add the JAVA_SOME variable with the value of the jdk installation address**



- **Confirm all the way until all the windows are closed**

Tutorial with Project Examples

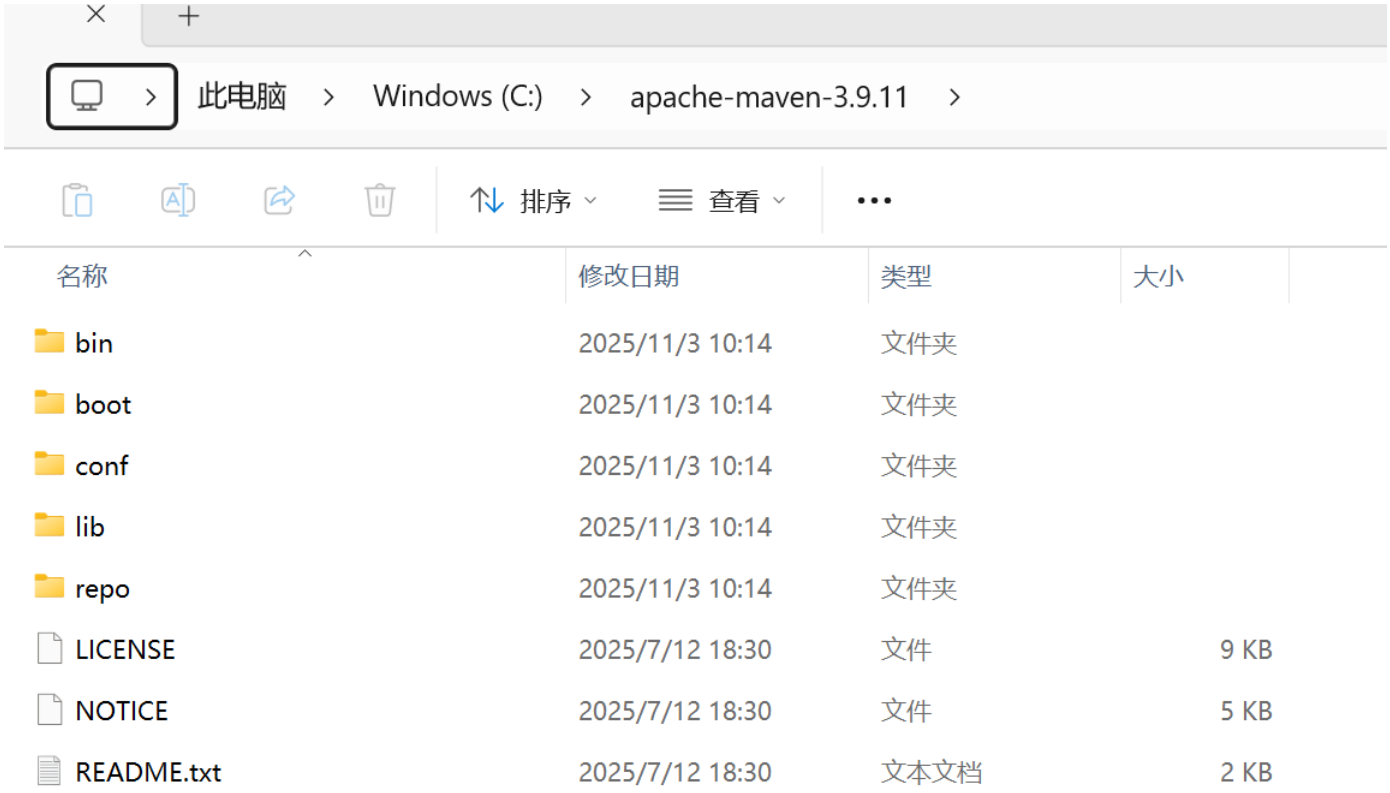
- Verify installation
- Press Win + R, enter cmd
- java -version
- javac -version



Tutorial with Project Examples

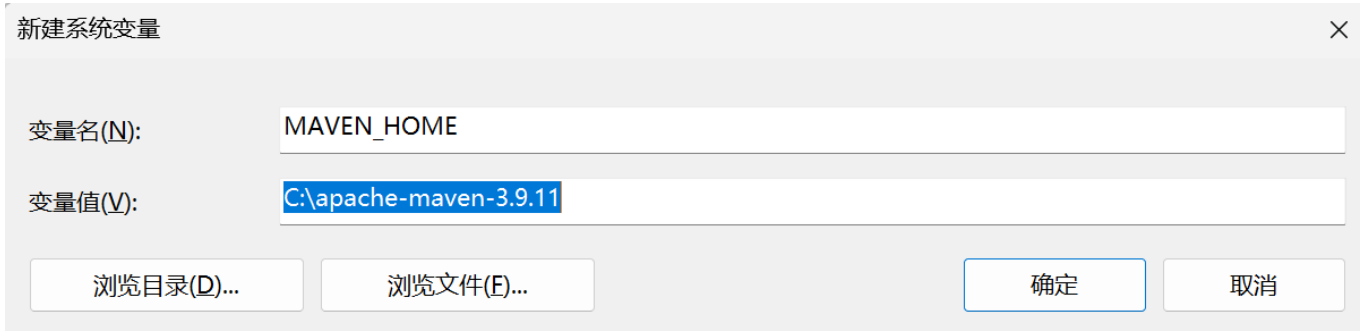
Download Apache Maven – Maven

- **Extract the downloaded Maven compressed file**
- **Enter the folder and create a repo folder to store Maven local repository dependencies**

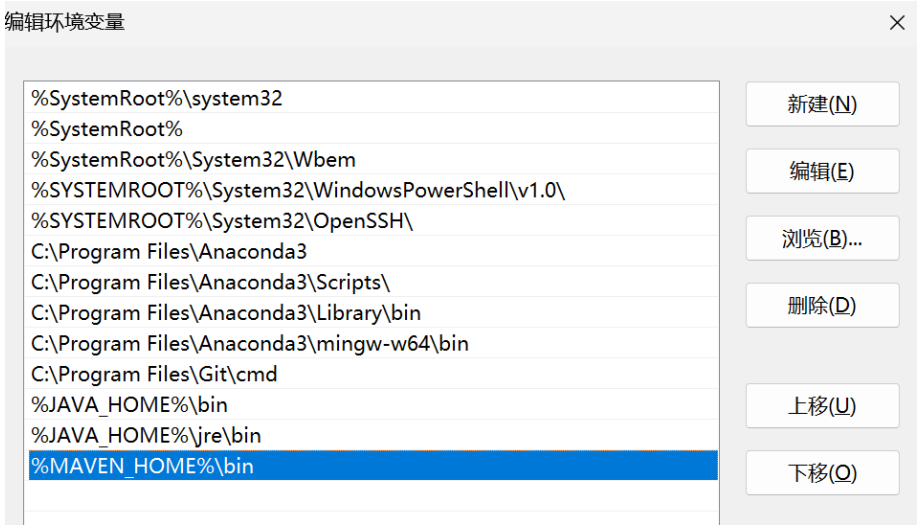


Tutorial with Project Examples

- **Create a new system variable MAVEN_SOME with the value of the Maven installation address**

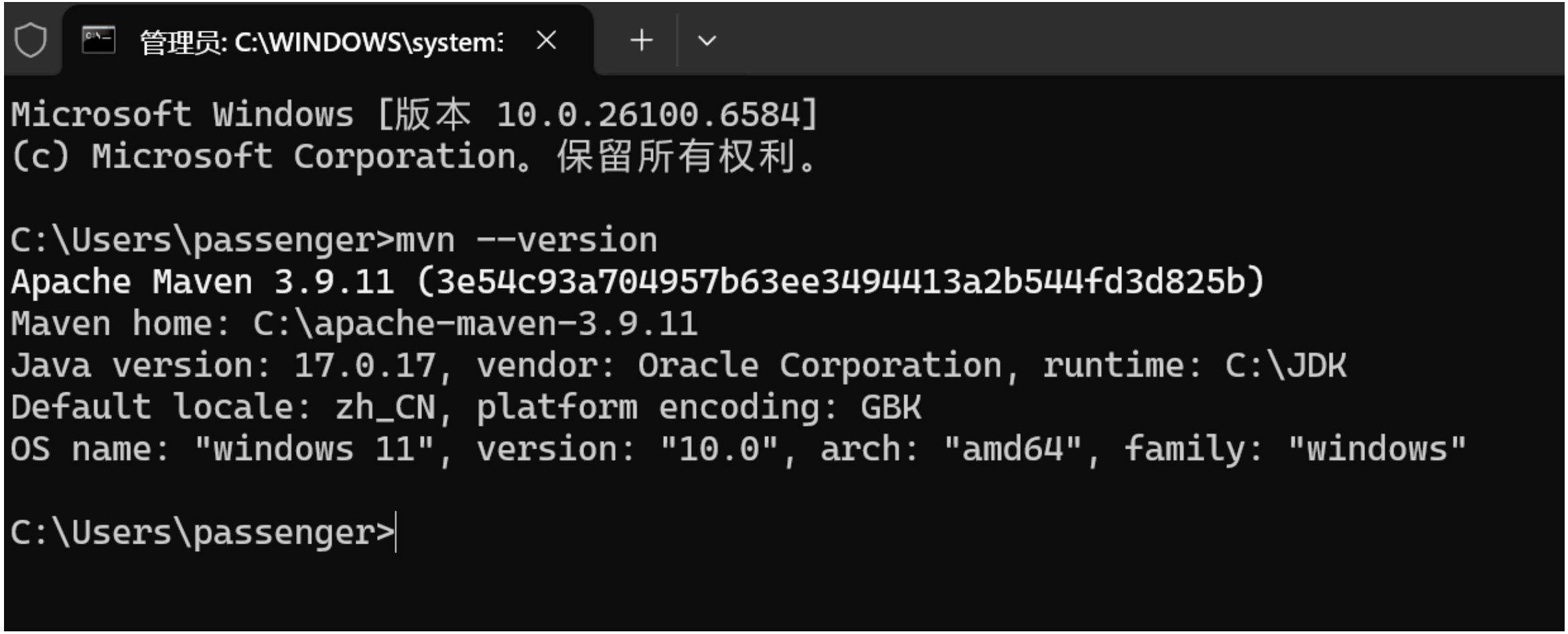


- **Create a new environment variable %MAVEN_HOME%\bin**



Tutorial with Project Examples

- **Win + R -> CMD， enter mvn --version (View Maven version)**



Tutorial with Project Examples

- Go to the config folder and edit the settings. xml file
- Specify repo address

```
<settings xmlns="http://maven.apache.org/SETTINGS/1.2.0"
          xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
          xsi:schemaLocation="http://maven.apache.org/SETTINGS/1.2.0 https://m
  <!-- localRepository
   | The path to the local repository maven will use to store artifacts.
   |
   | Default: ${user.home}/.m2/repository
  <localRepository>/path/to/local/repo</localRepository>
  -->
  <localRepository>C:\apache-maven-3.9.11\repo</localRepository>
```

- Configure mirrors

```
  <!-- 阿里云镜像 -->
<mirror>
  <id>aliyun-maven</id>
  <mirrorOf>central</mirrorOf>
  <url>https://maven.aliyun.com/repository/public</url>
  <blocked>false</blocked>
</mirror>
```

- Configure JDK version

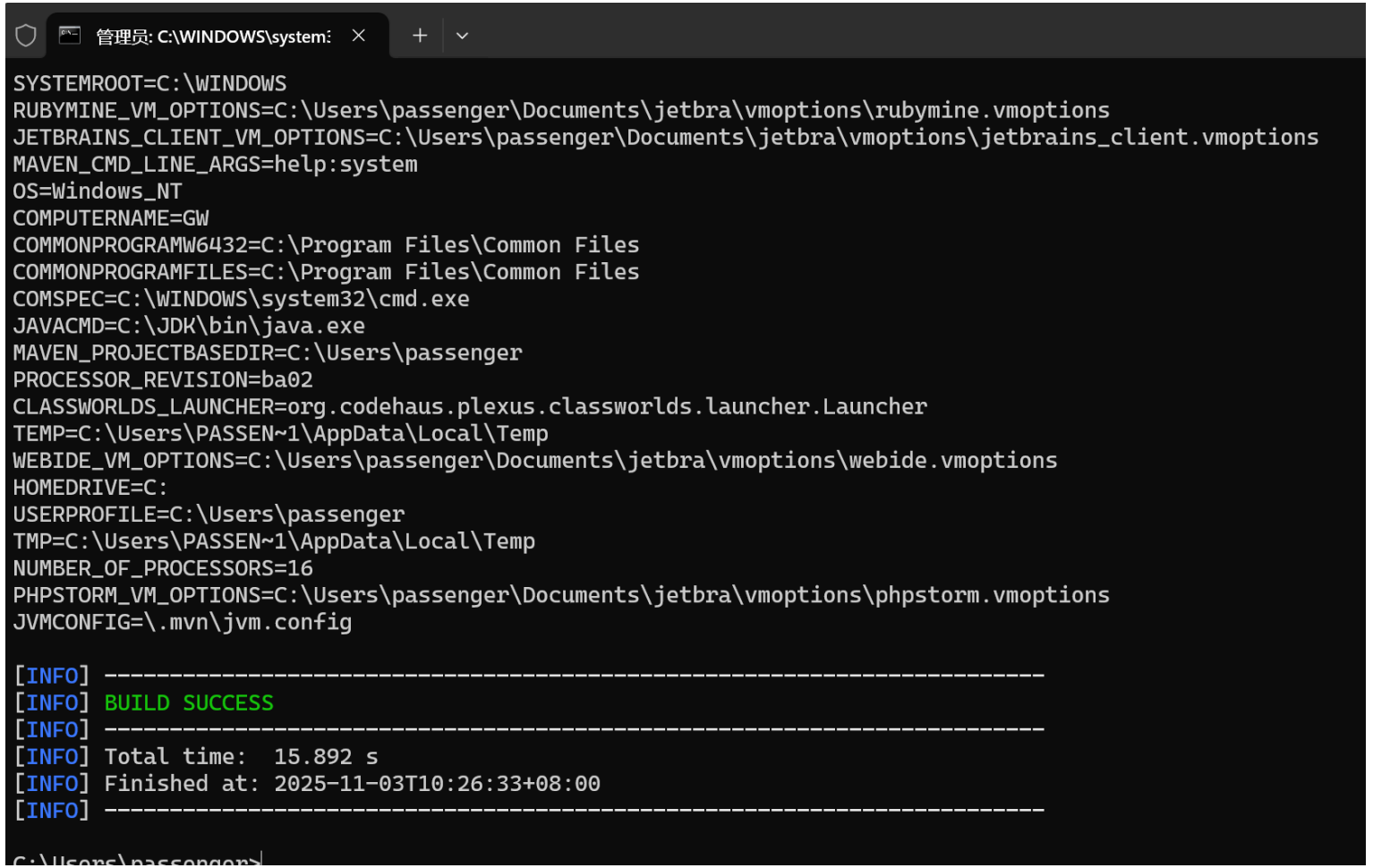
```
<!-- JDK17 -->
<profile>
    <id>jdk-17</id>
    <activation>
        <activeByDefault>true</activeByDefault>
        <jdk>17</jdk>
    </activation>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <maven.compiler.compilerVersion>17</maven.compiler.compilerVersion>
    </properties>
</profile>
```

The file sent to everyone is an edited version that does not require any changes except for the repo address

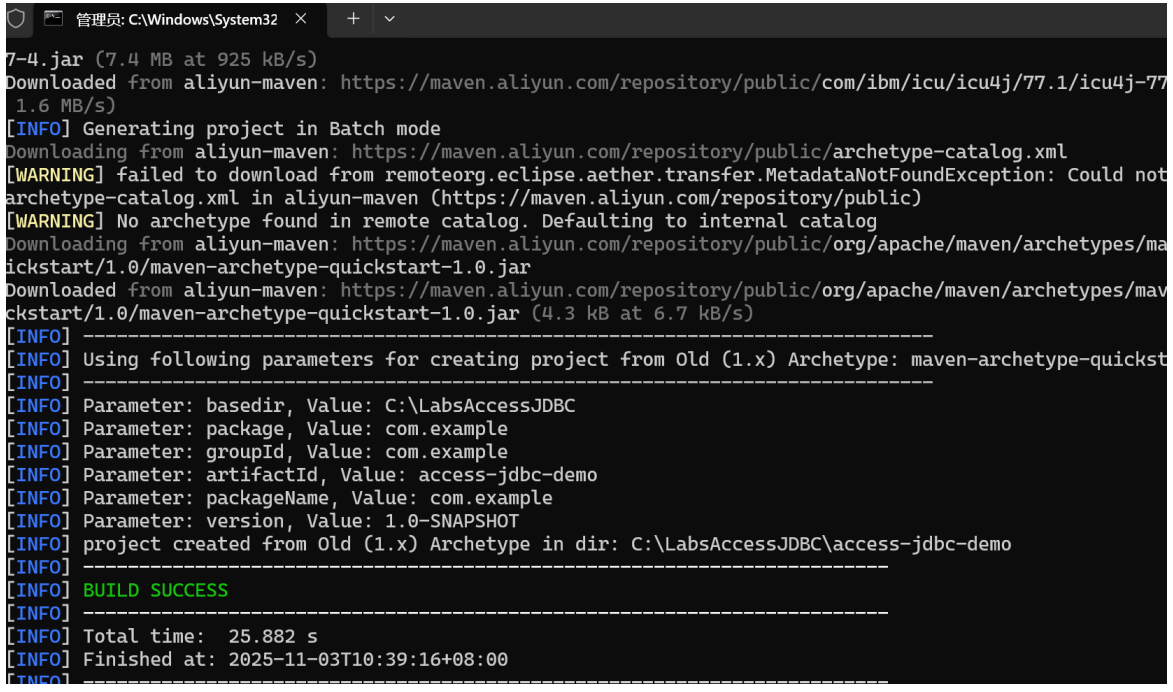
Tutorial with Project Examples

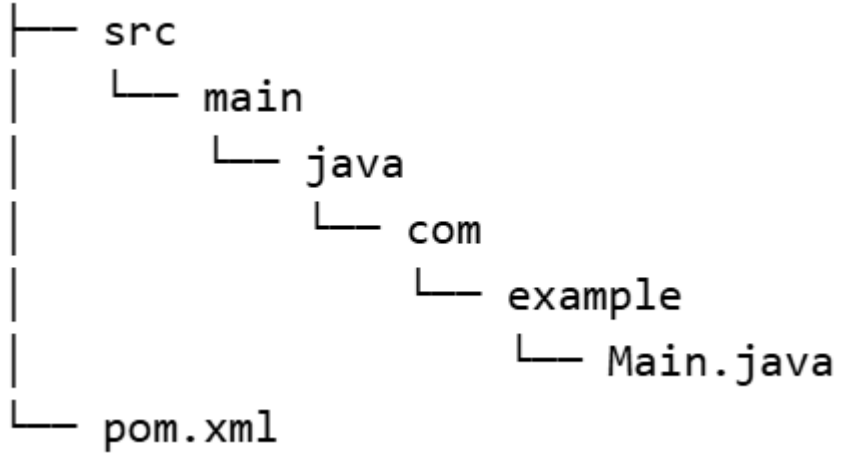
- **Check if the configuration is successful**
- **Win + R -> CMD , enter mvn help:system (Print Maven information)**



Tutorial with Project Examples

- Create a new folder named LabsAccess JDBC as the project address
- Enter the folder, enter cmd in the address bar, and open the command line
- Enter at the command line:
- mvn archetype:generate -DgroupId=com.example -DartifactId=access-jdbc-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false





Tutorial with Project Examples

- **Edit pom.xml**

```
<dependencies>
  <!-- UCanAccess JDBC Driver for Access -->
  <dependency>
      <groupId>net.sf.ucanaccess</groupId>
      <artifactId>ucanaccess</artifactId>
      <version>5.0.1</version>
  </dependency>
```

Tutorial with Project Examples

- **Operate the database by writing Java code**

「被至少2个不同水手预订过」的船只名称

bname
tiger
rose

「被至少2个不同水手预订过」的船只名称

```
SELECT bname
FROM Boats
WHERE bid IN (
    SELECT bid
    FROM (
        SELECT DISTINCT R.sid, R.bid
        FROM Reserves AS R
    ) AS DistinctReserves
    GROUP BY bid
    HAVING COUNT(*) >= 2
);
```

```
public class Main {
    private static final String DB_PATH = "C:\\Users\\passenger\\Documents\\project2.accdb";
    private static final String DB_URL = "jdbc:ucanaccess://" + DB_PATH;

    Run | Debug
    public static void main(String[] args) {
        try {
            // UCanAccess 驱动（Maven 已引入，无需手动加载，但保留更清晰）
            Class.forName(className: "net.ucanaccess.jdbc.UcanaccessDriver");

            Connection conn = DriverManager.getConnection(DB_URL);
            System.out.println(x: "✅ 成功连接 Access 数据库！");

            // 执行 CRUD
            selectTest(conn);
            conn.close();
            System.out.println(x: "🔌 连接已关闭。");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    private static void selectTest(Connection conn) throws SQLException {
        String sql = "SELECT bname\r\n" + //
                     "FROM Boats\r\n" + //
                     "WHERE bid IN (\r\n" + //
                     "    SELECT bid\r\n" + //
                     "    FROM (\r\n" + //
                     "        SELECT DISTINCT R.sid, R.bid\r\n" + //
                     "        FROM Reserves AS R\r\n" + //
                     "    ) AS DistinctReserves\r\n" + //
                     "    GROUP BY bid\r\n" + //
                     "    HAVING COUNT(*) >= 2\r\n" + //
                     ");\r\n" + //
                     "";
        try (Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery(sql)) {
            System.out.println(x: "\n--- 查询结果 ---");
            while (rs.next()) {
                System.out.printf(format: "姓名: %s \n",
                    rs.getString(columnLabel: "bname"));
            }
```

```
PS C:\LabsAccessJDBC\access-jdbc-demo>  c:;
rt=dt_socket,server=n,suspend=y,address=loca
' 'com.example.Main'
? 成功连接 Access 数据库！

--- 查询结果 ---
姓名: tiger
姓名: rose
? 连接已关闭。
PS C:\LabsAccessJDBC\access-jdbc-demo>
```

Project 3

I. Tutorial with Project Examples

II. Assignment

Assignment

1. Assignment Content

1. Connect to the universities database using Java

2. Implement adding, deleting, updating, and querying database tables using Java

Assignment

1. Assignment Content

3. Questions

This assignment requires everyone to exercise their subjective initiative:

1. Based on existing tables in the database, create a new table and design the fields and primary keys for the created table. The report needs to explain the reasons for establishing this table.
2. Delete the newly created table.
3. Insert a row of record in the student table.
4. Delete the newly inserted record.
5. Update a record in the student table.
6. Restore the modified records to their original state.
7. Execute all the SQL statements in Project 2.

Assignment

1. Assignment Content

4. Schema and Sample Data

- **The schema of the database is provided below (keys are red, field types are omitted):**
  - **student(sid, sname, sex, age, year, gpa)**
  - **dept(dname, numphds)**
  - **prof(pname, dname)**
  - **course(cno, cname, dname)**
  - **major(dname, sid)**
  - **section(dname, cno, sectno, pname)**
  - **enroll(sid, grade, dname, cno, sectno)**

Assignment

1. Assignment Content

5. Data Files

- **Download the following data files from WeChat Group(The imports you made in Project 1).**
  - ➢ **course.txt**
  - ➢ **dept.txt**
  - ➢ **enroll.txt**
  - ➢ **major.txt**
  - ➢ **prof.txt**
  - ➢ **section.txt**
  - ➢ **student.txt**

It is recommended to use the original student.txt for better communication and comparison of correctness among classmates.

Assignment

2. A Few Points to Note

- Pay attention to the address of the database connection, a single slash is an escape character.
- This is an **individual assignment** – no group submissions are allowed.
- Hand in a report which outlines your answers and the process by which you arrived at them.